

XRAY: Comprehensive Page Content Drafts

Below is the expanded, comprehensive content for each of the pages linked in the footer. This content is structured to provide a complete layout for each page of the "XRAY - Concept Prerequisite Diagnostic System" website.

Navigation

1. Home (/home)

Page Title: XRAY: Uncover the Root Cause of Student Confusion

Hero Section:

- **Headline:** Stop Guessing. Start Diagnosing with Precision.
- **Sub-headline:** When students struggle with advanced topics, the gap in their knowledge often lies hidden in foundational prerequisites. XRAY uses advanced AI, a comprehensive knowledge graph, and live probing questions to trace backwards and pinpoint exactly where learning broke down.
- **Primary Call to Action (CTA):** [Run a Free Diagnostic Scan]
- **Secondary CTA:** [Watch How It Works]

Section 1: The Problem with Traditional Tutoring

- **Header:** Compounding Confusion is the Enemy of Learning
- **Body:** Traditional tutoring often focuses on treating the symptoms—repeating the current topic until the student memorizes it. But if a student is struggling with Calculus because they never truly mastered Algebraic factorization, repeating Calculus won't help. XRAY identifies these foundational cracks before they cause the entire structure to collapse.

Section 2: How XRAY Works (Step-by-Step)

1. **Input:** The student describes what they are confused about in natural language.
2. **Mapping:** Our embedding engine maps their query to a specific node in our subject knowledge graph.
3. **Diagnosis:** The engine traces backwards, systematically testing prerequisites.
4. **Probing:** AI generates live, dynamic questions to confirm mastery or identify gaps at each step.
5. **Remediation:** XRAY provides targeted resources to fix the exact root cause discovered.

Testimonial / Social Proof (Placeholder):

"XRAY completely changed how we handle intervention. Instead of spending weeks trying to figure out why a student is failing, XRAY tells us exactly which prerequisite concept they missed from three years ago, in under 5 minutes." – Dr. Sarah Jenkins, Director of Curriculum

2. Features (/features)

Page Title: System Features & Capabilities

Hero Section:

- **Headline:** Precision Diagnostics Powered by Advanced AI
- **Sub-headline:** Explore the technology that makes XRAY the most accurate learning diagnostic tool available.

Feature Breakdown:

1. Dynamic Concept Knowledge Graphs

- **Overview:** At the core of XRAY is a robust, mathematically structured map of educational concepts.
- **Details:** We don't just list topics; we map the exact dependencies. If Node D requires Node B and C, and Node B requires Node A, the system understands this hierarchy perfectly. This Directed Acyclic Graph (DAG) approach ensures no prerequisite is ever overlooked.

2. Natural Language Embedding Search

- **Overview:** Students don't need to know the technical term for what they don't understand.
- **Details:** By leveraging vector embeddings, a student can say "I don't get why the numbers flip when I move them across the equals sign," and XRAY will accurately map this to "Properties of Equality" or "Inverse Operations" within the knowledge graph.

3. Algorithmic Diagnostic Engine

- **Overview:** XRAY acts like an expert detective tracing a suspect.
- **Details:** The engine performs a reverse traversal through the knowledge graph. It doesn't waste time testing unrelated concepts; it uses optimized pathfinding algorithms to test the most likely points of failure first, saving time and reducing student fatigue.

4. AI-Powered Probing & Grading

- **Overview:** We prove the gap exists before we try to fix it.
- **Details:** XRAY uses Generative AI to create unique, context-aware probing questions on the fly. When a student answers, the AI grades the free-text response, determining if they possess true mastery, partial understanding, or a complete misconception.

5. Targeted Remediation Pathways

- **Overview:** Diagnosis is only half the battle; fixing it is the other.
- **Details:** Once the root cause is isolated, XRAY automatically generates a custom remediation plan. This includes step-by-step AI explanations tailored to the student's specific misconception, followed by practice problems to cement the newly acquired understanding.

3. About (/about)

Page Title: About XRAY & Our Mission

Hero Section:

- **Headline:** The Next Evolution in Personalized Education
- **Sub-headline:** Bridging the gap between cognitive science and artificial intelligence.

Our Mission:

Our mission is to eliminate the educational debt that students accumulate over years of schooling. We believe that every student can master complex subjects if the foundational building blocks are properly secured.

The Origin Story:

XRAY (originally developed under the project name Prereq-Sleuth) was born out of frustration. Educators spend countless hours trying to remediate students, often relying on guesswork or standardized tests that only provide high-level scores, not actionable insights. Our founders realized that by combining the structural rigor of knowledge space theory with the conversational capabilities of modern Large Language Models (LLMs), they could automate the exact process an expert tutor uses to find a student's "missing link."

Our Core Values:

- **Precision over Volume:** We value finding the one right thing to teach, rather than teaching ten things in hopes that one sticks.
- **Transparency:** Our AI doesn't operate in a black box. The knowledge graph provides a clear, explainable path for every diagnosis.
- **Empowerment:** We build tools to augment educators, giving them the insights they need to do what they do best: inspire and teach.

Meet the Team (Placeholder):

- [Photo] **Jane Doe**, Lead Engineer
- [Photo] **John Smith**, Cognitive Scientist
- [Photo] **Alice Johnson**, Head of Product

Resources

4. Documentation (/documentation)

Page Title: XRAY Official Documentation

Hero Section:

- **Headline:** Welcome to the XRAY Docs
- **Sub-headline:** Everything you need to deploy, configure, and use the XRAY Diagnostic System.

Table of Contents / Navigation:

1. Getting Started

- **System Requirements:** Minimum specifications for hosting the XRAY backend.
- **Installation Guide:** Step-by-step instructions for setting up the Python/FastAPI backend and Vanilla JS frontend.
- **Environment Variables:** Configuring your `.env` file, including `GEMINI_API_KEY` and database paths.

2. User Guide (For Educators)

- **Running a Scan:** How to input a student query and initiate the diagnostic process.
- **Reading the Graph:** Understanding the visual UI, mastery states (Mastered, Struggling, Unknown), and traversal paths.
- **Reviewing Remediation:** How to access and assign the generated remediation materials.

3. Administration

- **Database Management:** How XRAY uses SQLite (aiosqlite) to manage session data and mastery states.
- **Modifying the Knowledge Graph:** Instructions on how to add new subjects, nodes, and prerequisite links to the `graph.py` structure.

4. Troubleshooting & FAQ

- Common errors (e.g., API rate limits, database locks) and how to resolve them.

5. API (`/api`)

Page Title: XRAY Developer API Reference

Hero Section:

- **Headline:** Integrate XRAY into your LMS or EdTech Platform
- **Sub-headline:** Powerful REST endpoints to bring prerequisite tracing to your own applications.

Overview:

The XRAY API allows developers to programmatically interact with our Diagnostic Engine. You can submit queries, receive graph states, request probing questions, and grade answers—all via standard HTTP requests.

Authentication:

All API endpoints require a Bearer token in the Authorization header. (See [Authentication Guide] for details on obtaining a token).

Core Endpoints:

- `POST /api/diagnose`

- o **Description:** Initiates a diagnostic traversal based on a student's free-text query.
- o **Payload:** {"query": "I don't understand how to multiply matrices."}
- o **Response:** Returns the matched node ID and the initial backward traversal path.

- **GET /api/probe/next**

- o **Description:** Fetches the next AI-generated probing question to test a specific concept node in the traversal path.
- o **Parameters:** ?node_id=matrix_dimensions
- o **Response:** Returns a question string and relevant context.

- **POST /api/probe/answer**

- o **Description:** Submits a student's answer for AI grading. The AI determines if the node is mastered or if the student is struggling, which dictates the next traversal step.
- o **Payload:** {"node_id": "matrix_dimensions", "answer": "You multiply the rows by the columns."}
- o **Response:** Returns a score, feedback, and the next_action (e.g., test the next prerequisite, or stop and remediate).

- **GET /api/remediation/{node_id}**

- o **Description:** Retrieves targeted explanations and practice questions for the identified root cause.

(Note: Download our full OpenAPI/Swagger specification file for complete schemas, response codes, and examples).

6. Research (/research)

Page Title: Research, Methodology, & Whitepapers

Hero Section:

- **Headline:** The Science Behind XRAY
- **Sub-headline:** Grounded in cognitive science, powered by machine learning.

Our Methodology:

XRAY's approach is built on the intersection of two distinct fields of study:

1. **Knowledge Space Theory (KST):**

First proposed in the 1980s by Doignon and Falmagne, KST suggests that a student's knowledge state can be modeled mathematically. By defining strict prerequisite relationships as a Directed Acyclic Graph (DAG), we create a deterministic map of what a student *must* know before learning a new concept. XRAY brings this theory into practice through its rigid graph structure.

2. **Large Language Models (LLMs) & Natural Language Processing (NLP):**

While KST provides the map, LLMs provide the vehicle. Traditional KST systems rely on massive

banks of pre-written multiple-choice questions. XRAY uses advanced NLP (via embedding vectors) to map messy, real-world student queries onto the strict mathematical graph. Furthermore, we use Generative AI to dynamically create probing questions and grade free-text answers, simulating a Socratic dialogue.

Publications & Whitepapers:

- **Whitepaper:** *Tracing the Missing Link: Automated Prerequisite Diagnostics using DAGs and Generative AI.* (Download PDF)
 - **Case Study:** *Reducing Remediation Time by 40%: A Pilot Study of XRAY in University Linear Algebra Courses.* (Read Online)
 - **Technical Architecture:** *Building Scalable Diagnostic Engines with FastAPI and Vector Embeddings.* (Read Online)
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Connect

7. Connect (/connect or /contact)

Page Title: Connect with the XRAY Team

Hero Section:

- **Headline:** Let's Talk Diagnostics
- **Sub-headline:** Whether you're an educator, developer, or researcher, we want to hear from you.

Contact Information:

- **General Inquiries:** hello@xray-diagnostics.edu
- **Technical Support:** support@xray-diagnostics.edu
- **Partnerships:** partnerships@xray-diagnostics.edu

Join the Community:

- **GitHub:** [View the Prereq-Sleuth / XRAY Repository] - Contribute to the open-source core!
- **Discord:** [Join our Developer & Educator Discord] - Discuss integrations, graph design, and AI prompts.
- **Twitter/X:** [@XRAY_Diagnostics]

Send us a Message:

Use the form below to get in touch directly. Our team typically responds within 24 hours.

- **Name:** [_____]
- **Email:** [_____]
- **I am a:** [Dropdown: Educator / Developer / Researcher / Student / Other]
- **Message:** [_____]
- [Submit Message] button